



OBJECT ORIENTED SOFTWARE ENGINEERING

Ms. Archana A

Department of Computer Applications

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DevOps Approach

Archana A

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DevOps, at its core, is a philosophy that aims to **bridge the gap** between **software development (Dev)** and **IT operations (Ops) teams**.

It is a set of practices, tools, and cultural changes that foster **collaboration, automation, and continuous improvement** throughout the entire software development lifecycle.

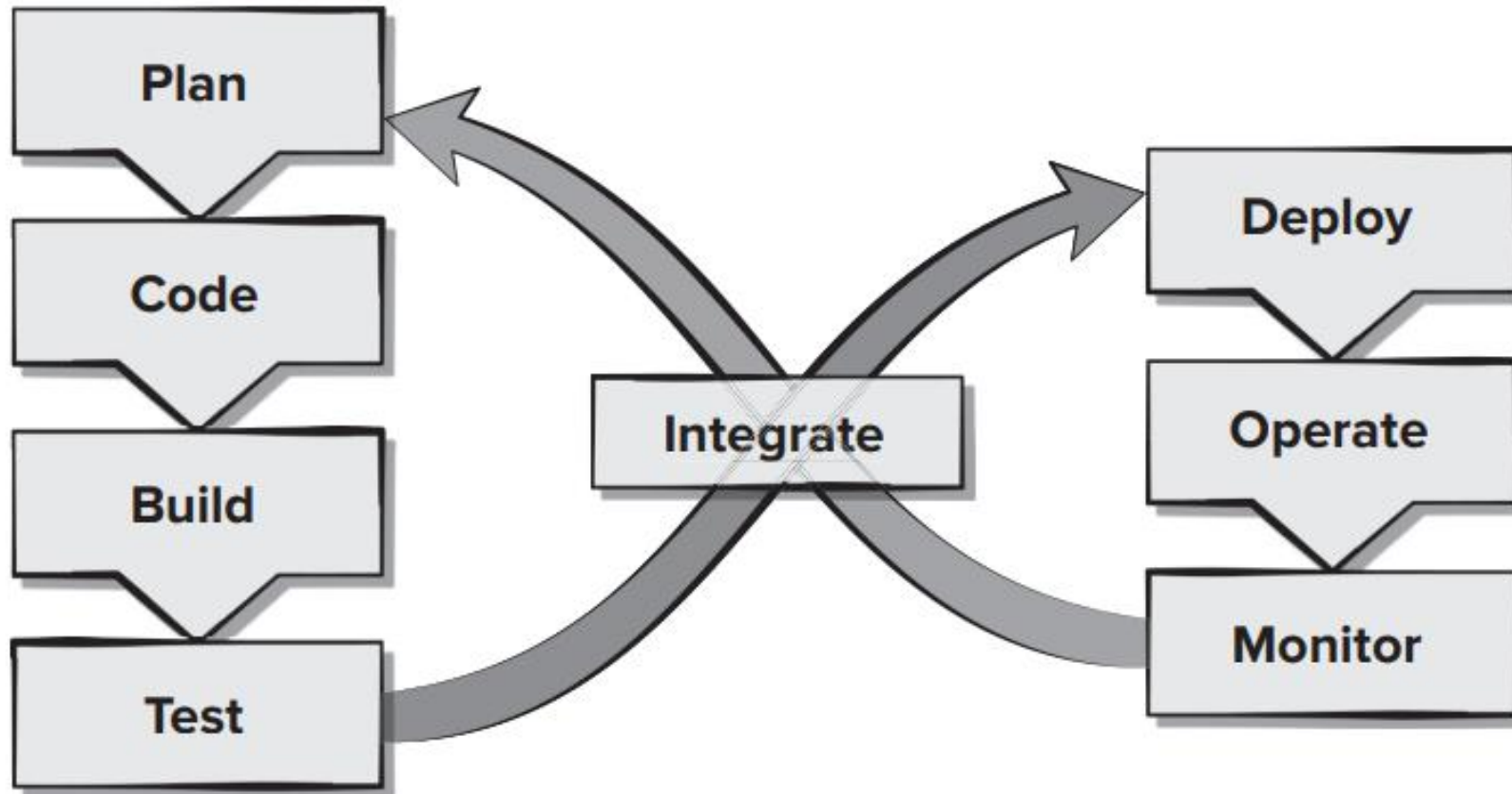
Working of DevOps: consider an example

Imagine a bakery:

- **Traditionally (without DevOps):** The bakers (Dev) create new recipes and bake delicious treats. But getting those treats to the customers (production) is a separate process handled by the delivery team (Ops). **Communication is limited, leading to delays, confusion, and sometimes even stale treats reaching the customers.**
- **With DevOps:** The bakers and delivery team **work together**. They share recipes, understand each other's challenges, and automate tasks like mixing ingredients and packaging. This allows them to rapidly experiment with new recipes, identify issues early, and deliver fresh treats to customers faster and more reliably.

- DevOps was created by Patrick DeBois to **combine Development and Operations**.
- DevOps attempts to apply **agile and lean development** principles across the entire software supply chain.
- Figure (next slide) presents an **overview of the DevOps workflow**.

DevOps



- The DevOps approach involves **several stages** that loop continuously until the desired product exists:

1. Continuous development. Software deliverables are broken down and developed in multiple sprints with the increments delivered to the **quality assurance members** of the development team for testing

2. Continuous testing. Automated testing tools are used to help team members test multiple code increments at the same time to ensure **they are free of defects prior to integration**

3. Continuous integration. The code pieces with **new functionality** are added to **the existing code** and to the run-time environment and then checked to **ensure there are no errors** after deployment.

4. Continuous deployment. At this stage the integrated code is deployed (installed) to the production environment, which might include multiple sites globally that need to be prepared to receive the new functionality.

5. Continuous monitoring. Operations staff who are members of the development team help to **improve software quality** by **monitoring its performance** in the production environment and proactively looking for possible problems before users find them

Key DevOps principles:

- **Automation:** Repetitive tasks like infrastructure provisioning, testing, and deployment are automated, freeing up time for more creative and strategic work.
- **Continuous integration and continuous delivery (CI/CD):** Changes are integrated and deployed frequently, allowing for faster feedback and quicker fixes.
- **Shared responsibility:** Both Dev and Ops teams take ownership of the entire software lifecycle, from development to production.
- **Monitoring and feedback:** Continuous monitoring and feedback loops help identify and address issues quickly.

Benefits of DevOps:

- **Faster software delivery:** Get new features and updates to customers quicker.
- **Improved quality:** Catch and fix bugs early in the development process.
- **Increased reliability:** Fewer production issues and downtime.
- **Better team collaboration:** Improved communication and shared goals between Dev and Ops.

Note: Remember, DevOps is not just a set of tools, it's a cultural shift. It requires a change in mindset and collaboration to truly reap its benefits.



THANK YOU

Ms. Archana A

Department of Computer Applications

archana@pes.edu

+91 80 6666 3333 Extn 392